Plan and Milestones

	Activities	**Approaches**	**Tools**
1	Background reading	Read textbook chapters and academic papers, consolidating information in to a paragraph summarising the basics of each topic	University Online Library
2	Reading academic papers of existing models specific to similar problems	Use a key paper to find relevant papers through citations. Keep track of papers and useful quotes	Chenarlogh et. al, 2022 & Notion databases
3	Reading papers of pertinent architectures	In particular regarding U-Net, TransUNet and Vision Transformers	Scopus & Google Scholar
4	Submit Ethics Approval Form	Fill in and check with Jon	Ergo
5	Literature Review	Use filled Notion Database to write, referencing as I go	BibTeX referencing
6	Load and preprocess data	Load data, then preprocess data e.g. data augmentation to simulate minor rotation and speckle changes	NumPy, e.g. MONAI
7	Statistical analysis on the dataset	Find distributions of abdominal size, number of scans per patient, contextualise information 'baked-in' to the images. Use this to determine sensible train/test splits	Meaure region properties with scikit-image
8	Benchmark results	Apply pre-trained off-the-shelf model to test set and calculate various evaluation metrics for comparison later	Meta AI's SegmentAny-thing model
9	Finalise the planned project approach and write planned approach in report	Define the overall progression of the project and methodologies for implementation	Detailed Gantt Chart
10	Implement/ fine tune U-Net	Build or adapt a U-Net model, tuning the parameters for optimal performance on this dataset	PyTorch & Iridis
11	Modify U-Net for performance improvement	Integrate changes to the architecture, taken from ideas implemented in academic papers read in the literature review	Physical development log
12	Evaluate accuracy of segmentation using useful metrics	Use a range of region and distance metrics to thoroughly assess model performance	Dice similarity, intersection over union, absolute diff, Hausdorff dist
13	Reflect on project progress and compare accuracy achieved with similar papers	Consider accuracy of model achieved so far, considering the best implementation route forward	PERT and Gantt Chart
14	Consider vision transformers and TransUnet	Consider implementing a transformer layer in the UNet to improve global context capture and compare model with U-Net alone	Feature Attribution